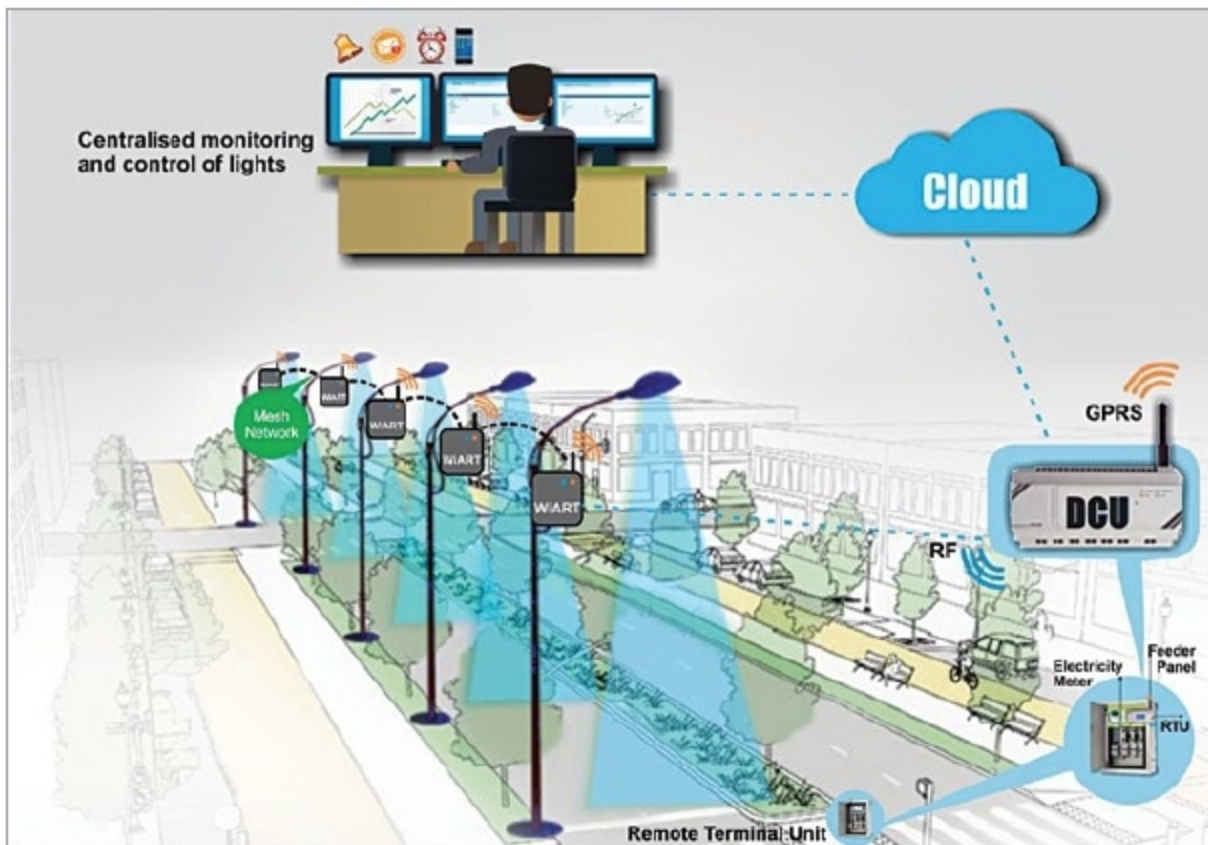




ensure optimum utilisation of existing resources.

Maven Systems was founded in Pune, in 2011. It provides smart solutions to optimise energy efficiency and productivity on a large scale. Its solutions allow monitoring of fuel consumption in various sectors such as excavators, loaders, trucks, containers and diesel generators. Monitoring fuel enables users to ensure its optimum utilisation and get immediately notified on geofencing violations to prevent theft.

There are various other wireless solutions such as smart metering, smart sub-metering (for individual households) and smart lighting solutions that allow users to access consumption data and control devices

from anywhere. These solutions can significantly improve energy efficiency.

Their automated streetlighting solutions allow users to control multiple streetlights from a single device. The solution combines multiple lights into one phase or group, which can be controlled through a single panel and provide reports for the entire panel. Switching on and switching off streetlights can be scheduled, monitored and changed via SMS. Detailed consumption reports are also available. Maven Systems provides a tailored solution to match such requirements as light intensity, which, in turn, can reduce electricity consumption by 30 per cent.

Another energy-monitoring solution is township monitoring, which is

a combination of utility monitoring, diesel generator set monitoring and periphery lighting. Users can monitor operational hours, energy produced and fuel consumption. Data is uploaded to the cloud application in real time and can be accessed using a smart-phone. The solution also includes fuel sensors that provide fuel data with 98 per cent accuracy, which can help users identify exact consumption of fuel and prevent theft.

Maven's end-to-end smart metering solutions do not require any hardware, protocol changes or additional cost for wiring. Data-collection procedures include data concentrator units and handheld units. The solutions are integrated with multiple meters that meet global standards.

The solutions are based on machine-to-machine (M2M) and Internet of Things (IoT) technologies. These deploy sensors to conventional devices, which deliver data to the cloud through a gateway that can be accessed remotely.

The solutions benefit industries that are providing services in industrial automation, home automation, smart metering, building management systems and lighting control.

Maven Systems has deployed their solutions in 28 countries and achieved a place in "Top 10 IoT companies in India" listed by NASSCOM and Business World. Undoubtedly, such solutions are imperative to sustainably run businesses and will eventually result in enhanced profitability.

—Deeksha Sharma

4 Innovation For Monitoring IONOSPHERIC DISTURBANCES

VLF SID monitor is a very-low-frequency (VLF) sudden-ionospheric-disturbance monitor. It is a radio receiver operating in the frequency range of 3kHz to 30kHz VLF band of the electromagnetic spectrum. Most broadcast receivers give out voice or music output, or digital

data for modern-day communication receivers, whereas VLF SID monitors do not. These receive man-made or natural radio signals in VLF band. Here, there is no demodulation or extraction of data or voice contents—the receiver simply provides a calibrated DC voltage output as a function

of input RF signal strength.

This receiver finds application in remote sensing and scientific studies related to the ionosphere. Ionosphere is a result of ionisation of Earth's upper atmosphere, primarily due to the Sun and solar radiations. VLF radio/electromagnetic signals travel